



Prepared by group 21

CISC_{351/372} Group Project

Multi-perspective analysis of the Amazon Reviews dataset

15 April, 2025



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Introduction

- Online reviews are critical to modern e-commerce.
 - Star ratings (quantitative)
 - Written text (qualitative)
- However, these two signals often **contradict** each other.
 - high ★ + negative comments
 - low ★ + 👍
- This inconsistency — called **sentiment-rating misalignment** — raises concerns:
 - Misleads consumers and businesses
 - Challenges sentiment analysis and recommendation models
 - Affects product and brand reputation
- This project addresses three key questions:
 - RQ1: How often do textual sentiment and star ratings differ?
 - RQ2: What factors cause these mismatches?
 - RQ3: How do such reviews influence product reputation over time?



Motivation

- Star ratings and review texts are widely used as feedback signals in e-commerce.
- However, platforms often assume they convey the same user opinion — which is not always true.
- Sentiment-rating misalignment:
 - Misleads consumers about product quality
 - Undermines the reliability of recommendation systems
 - Reduces trust in brands and platforms
- Most existing systems ignore this discrepancy.
- By investigating this problem, we aim to:
 - Improve the credibility of review systems
 - Enhance sentiment analysis models
 - Understand user behavior and bias in review generation



Dataset Statistics

- Our dataset consists of Amazon product reviews, each containing:
 - **Star rating**: given by the user, 1-5 ★
 - **Written review** text describing the user's experience
 - **Predicted sentiment score**: generated using a fine-tuned BERT model
 - **Verified purchase flag**: indicating if the user bought the item via Amazon
 - **Helpful vote**: count showing how many users found the review useful
 - **Timestamp**: when the review was posted
 - **Product ID (ASIN)&parent ASIN**: group product variants
 - **User ID**: representing the reviewer identity
 - **Review title** (optional but often summarizing key sentiment)
- After preprocessing, we retained a compact but informative set of fields to capture:
 - **Emotional expression** (via text and BERT sentiment)
 - **Behavioral context** (verified purchase, helpfulness)
 - **Structural attributes** (product and user metadata)



Dataset Statistics - Examples

Field	Example 1	Example 2
Rating	3.0	4.0
Title	"Meh...."	"Too Small, No Pockets"
Text	"It's ok... didn't match... bottoms were cut weird..."	"No pockets... soft and comfortable... size up..."
Predict Stars	2	3
Verified Purchase	1	1
Helpful Vote	0	3
Timestamp	1593109292435	1668894486799
ASIN / parent	B07THQ121H / B07DYJVTTN	B01N6EBIII / B06VY7R1MZ
User ID	AEQIVU6BF23G3ELOJDCE55WMERUA	AEUZGUNOA6YN6F5EBRMHV7K6OMVA



RQ1 Methodology Overview



Data Preparation:

We selected 1.5 million reviews from 3 categories.

Preprocessing steps:

- Remove empty/missing data
- Lowercase + clean text
- Drop irrelevant fields (e.g., images)

Sentiment Prediction:

We compared 3 models:

- **VADER** (lexicon-based)
- **BERT**
- **RoBERTa**

We selected BERT for its balanced F1 score and better interpretability.

Each review received a predicted sentiment score (1–5).

RQ1: What is the difference between the emotional intensity of user reviews and the assigned ratings?



Discrepancy & Topic Modeling:

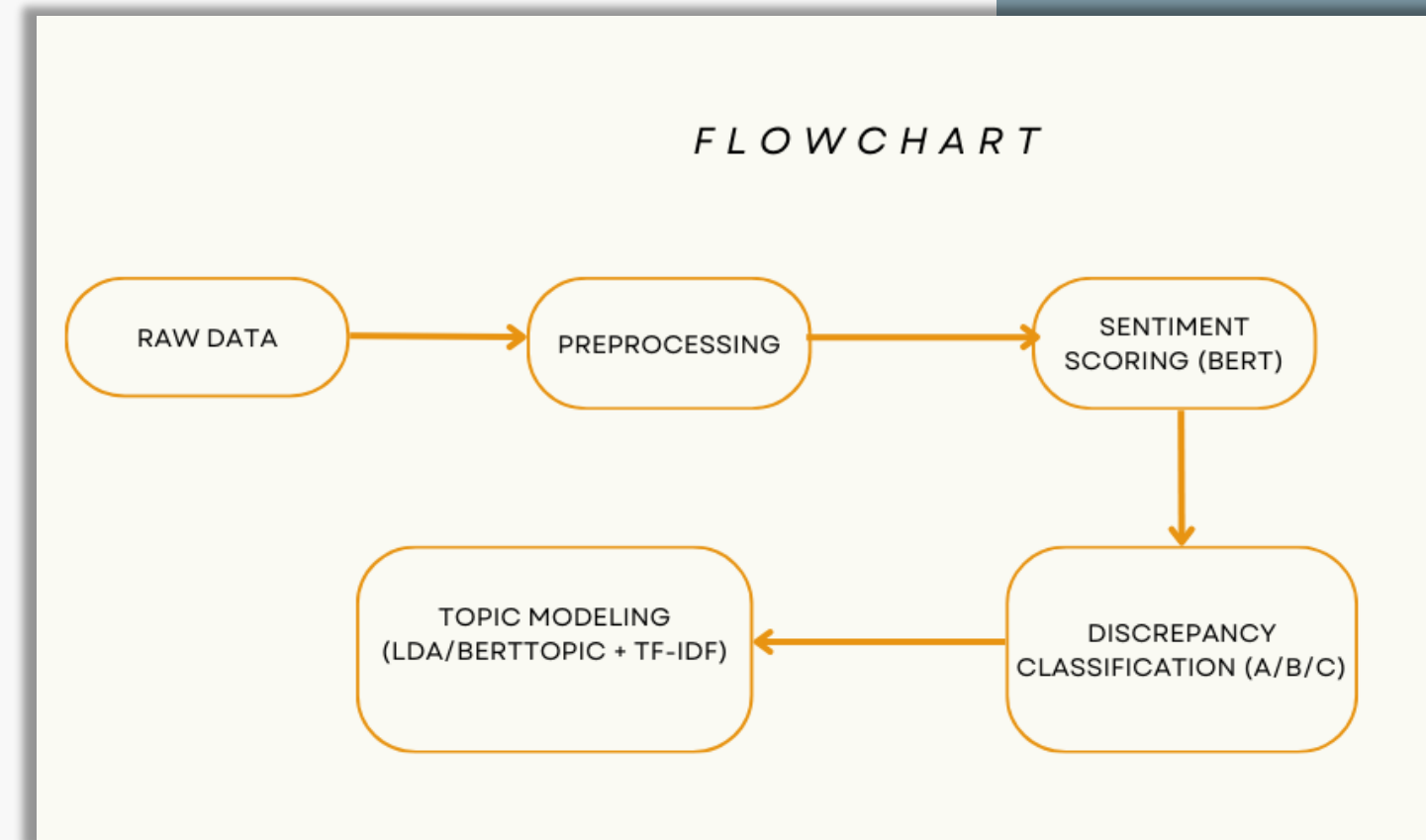
- Compute $\text{rating_gap} = \text{rating} - \text{predicted_score}$
- Categorize into:
- **Group A** (high rating, low sentiment)
- **Group B** (low rating, high sentiment)
- **Group C** (aligned)
- Apply **LDA + BERTTopic** to Group A & B for deeper insights.



RQ1 Methodology

Topic Modeling (Group A & B Only)

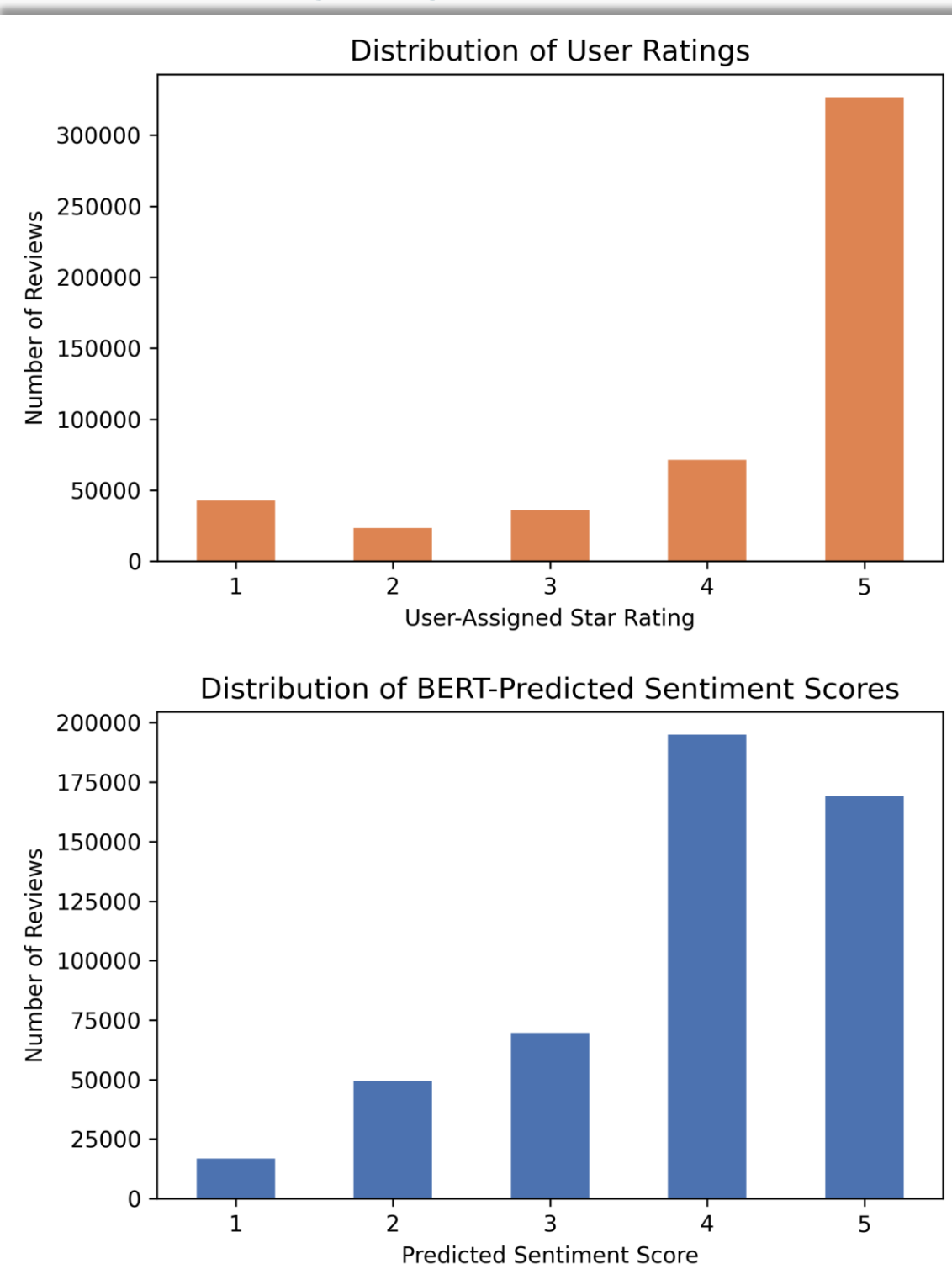
- Used **LDA** to identify thematic word clusters
- Used **BERTTopic** for contextualized sentence-level topic extraction
- Applied **TF-IDF** for high-signal keywords per group



RQ1 Experiments – Analysis Setup

Sentiment & Rating Distribution :

- Dataset: 3 product categories × 500,000 reviews each
- For each review:
 - predict_stars** from BERT
 - user rating** (original 1–5 star)

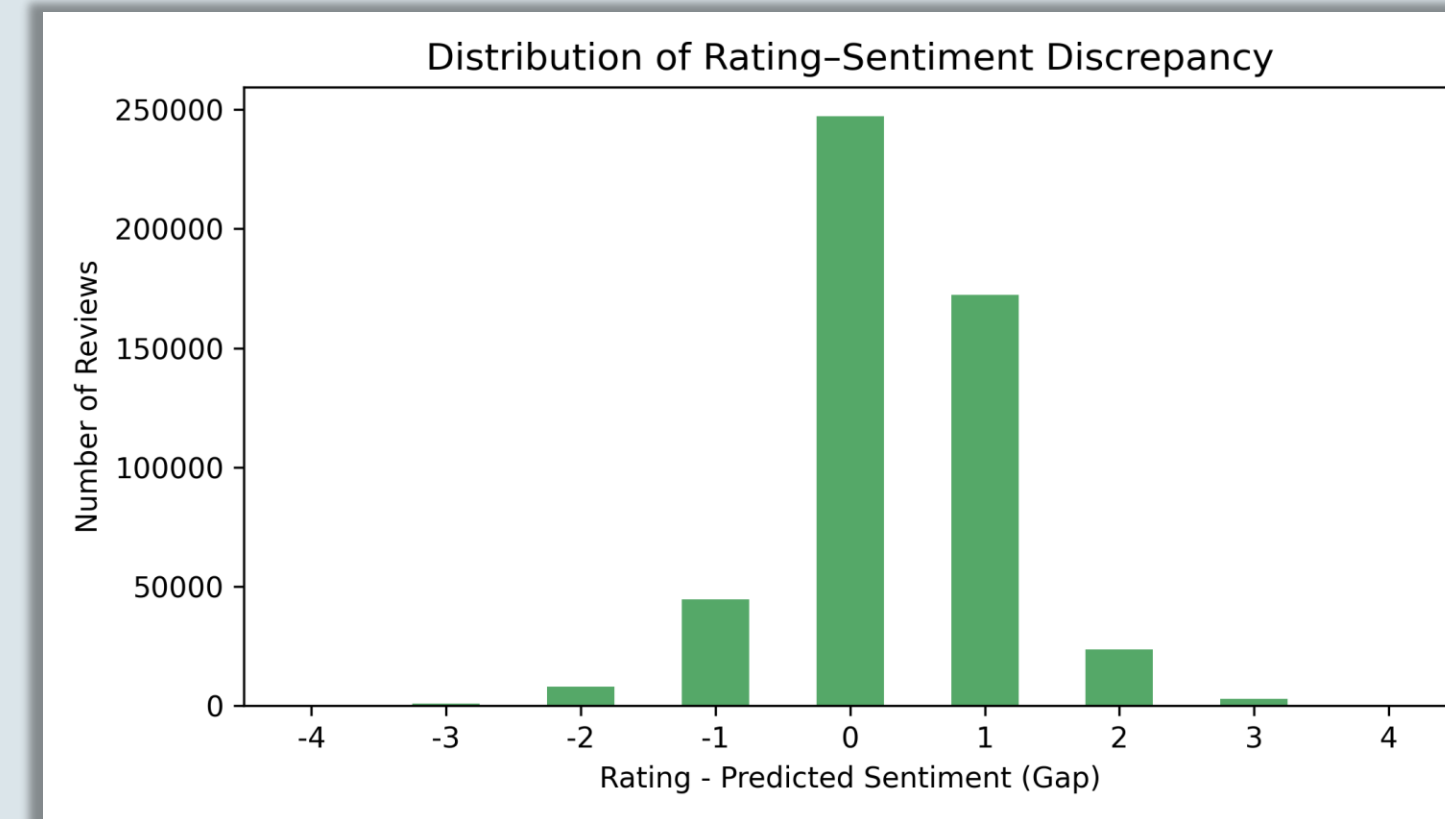


Explanation:

- Sentiment scores cluster around 4 → generally positive tone
- User ratings skew more heavily toward 5 → **positivity bias**

Rating Gap Computation:

- Formula: **rating_gap = rating - predict_stars**
- Distribution shows misalignment patterns across reviews



Observation:

- Group A** is the most dominant cluster
- Users frequently assign high scores even when expressing mild dissatisfaction



RQ1 Results – Keywords & Review Examples

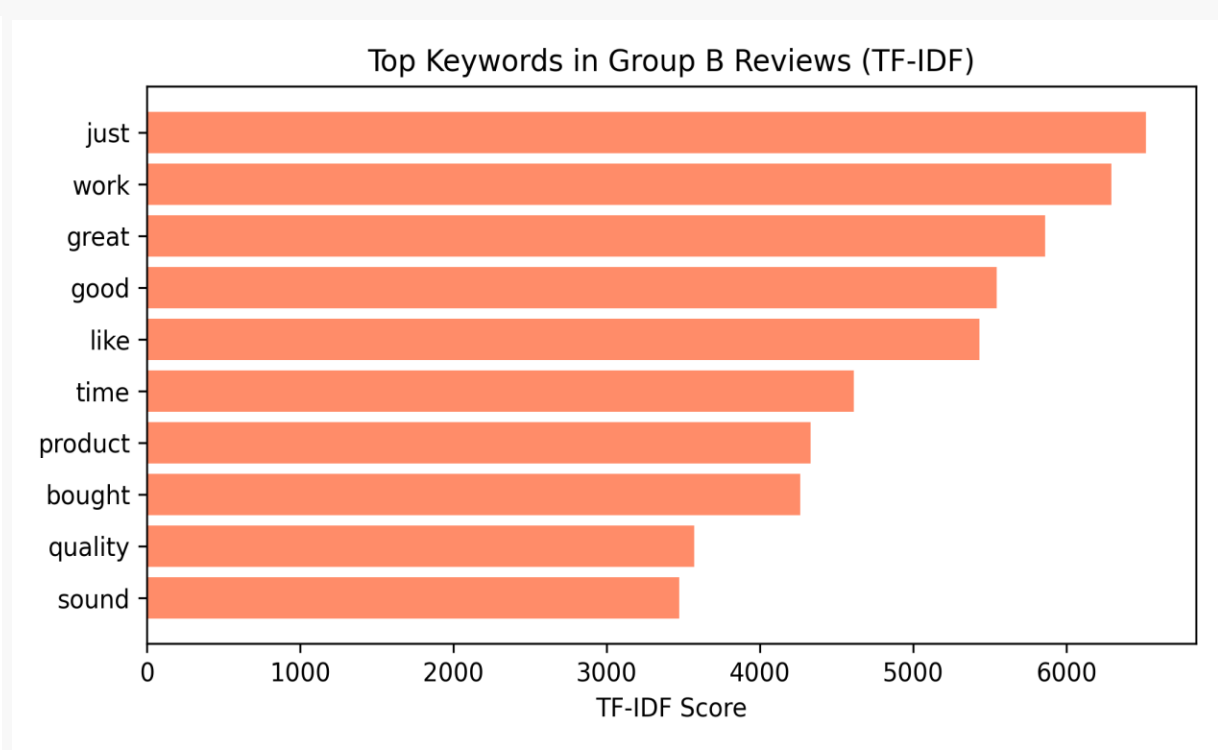
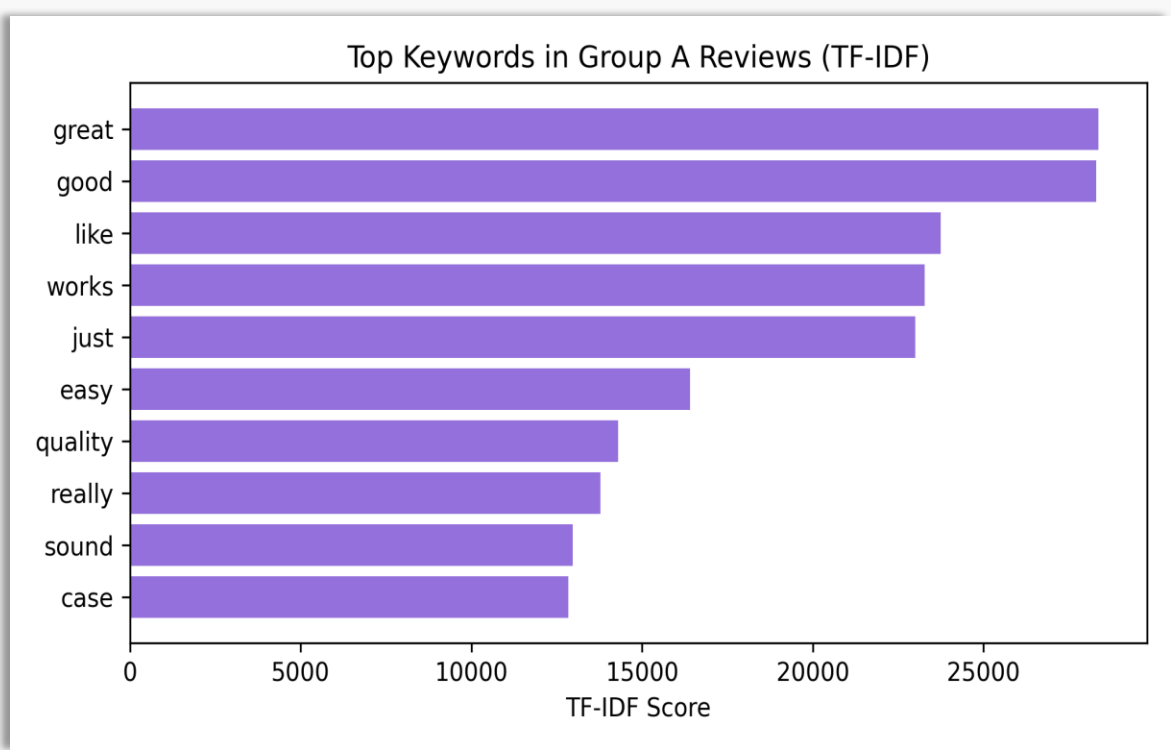


TABLE 1
Example reviews for each rating–sentiment discrepancy group.

Group	Rating	Sentiment	Review Excerpt
A	5	2	"Arrived late and damaged, but I like the look."
B	1	4	"Nice quality and works well, but shipping was slow."
C	4	4	"Exactly as described with fast delivery."

TF-IDF Keywords per Group

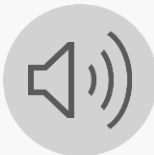
- **Group A:** "works", "easy", "sound"
- Functional focus despite neutral/negative tone

TF-IDF Keywords per Group

- **Group B:** "time", "bought", "just"
- Reflect frustration despite overall product positivity

Conclusion:

- These examples reveal the complexity of user intention
- Sentiment-rating divergence is influenced by politeness, expectations, and external factors



RQ₁ – Discussion



Key Observations

- **Key Observations**
- **Sentiment and star ratings often diverge**
- High ratings may be influenced by:
 - Brand loyalty
 - Politeness or social pressure
- **Group A is most frequent** — reviews with mild complaints but high scores

Challenges & Limitations:

- **Sarcasm and nuance** are difficult to capture even for advanced models like BERT
- Ratings may not always reflect review intent — some users rate high to avoid conflict or due to habit
- Cultural and contextual differences affect both emotional expression and rating behavior
- **BERT prediction**, while powerful, may still misinterpret subtle emotional tones

Implications:

- Sole reliance on star ratings can **mislead sentiment-based analysis and recommendation systems**
- Incorporating **textual sentiment insights** adds depth and accuracy to trust modeling and product evaluation
- Our classification framework (Groups A, B, C) can support **more transparent review platforms**



RQ2-1 Use TF-IDF vectorization and KMeans clustering (k=5)

 Top Keywords per Topic:

Topic 1: great, nice, good, fit, like, size, quality, cute, small, perfect

Topic 2: br, like, size, nice, fit, good, wear, just, comfortable, great

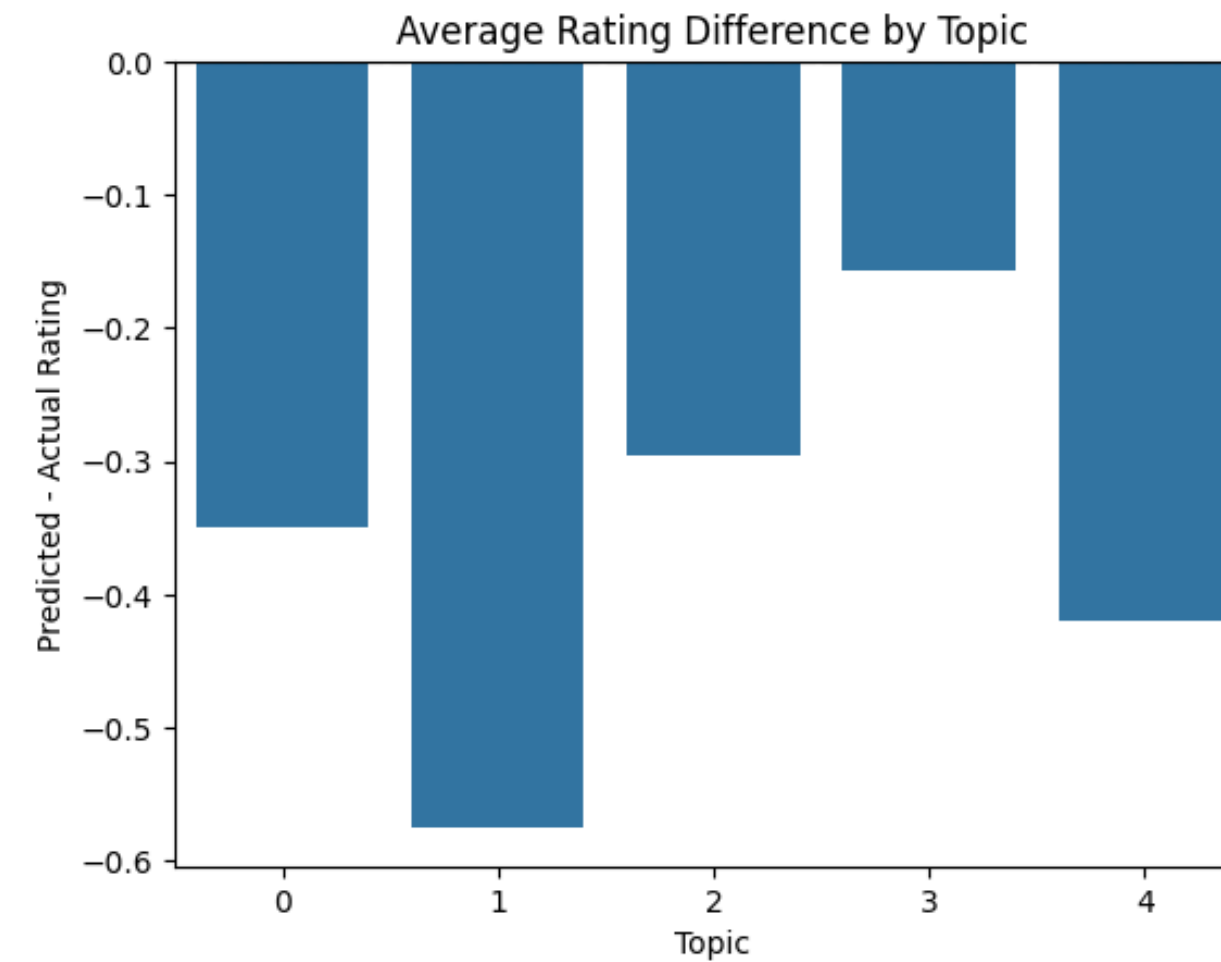
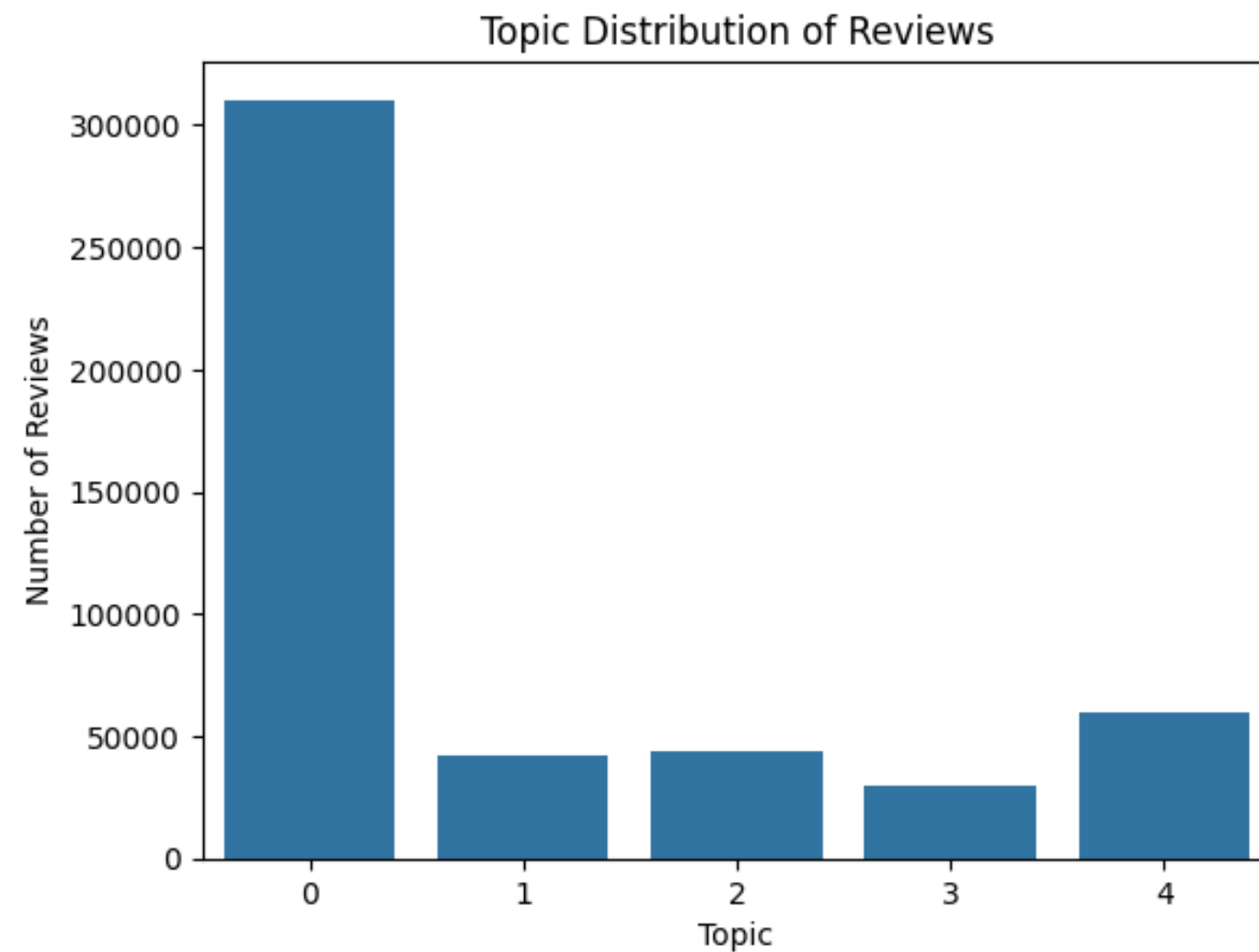
Topic 3: love, great, perfect, fit, comfortable, cute, soft, color, fits, shirt

Topic 4: stars, great, love, nice, comfortable, good, fit, quality, loves, cute

Topic 5: comfortable, shoes, great, fit, wear, shoe, good, soft, feet, nice



RQ2-1 Use TF-IDF vectorization and KMeans clustering ($k=5$)



RQ2-1 Use TF-IDF vectorization and KMeans clustering (k=5)



Text Length Distribution per Topic (%) :

length_category	Very Short	Short	Medium	Long	Very Long
topic					
0	39.375288	32.647771	19.959118	7.152783	0.865040
1	6.111363	14.643444	34.442830	29.634748	15.167616
2	54.422329	34.587051	9.064287	1.814230	0.112102
3	98.628720	1.111037	0.226878	0.033364	0.000000
4	38.713586	35.256896	18.649972	6.538093	0.841453



Verified Purchase Distribution per Topic (%) :

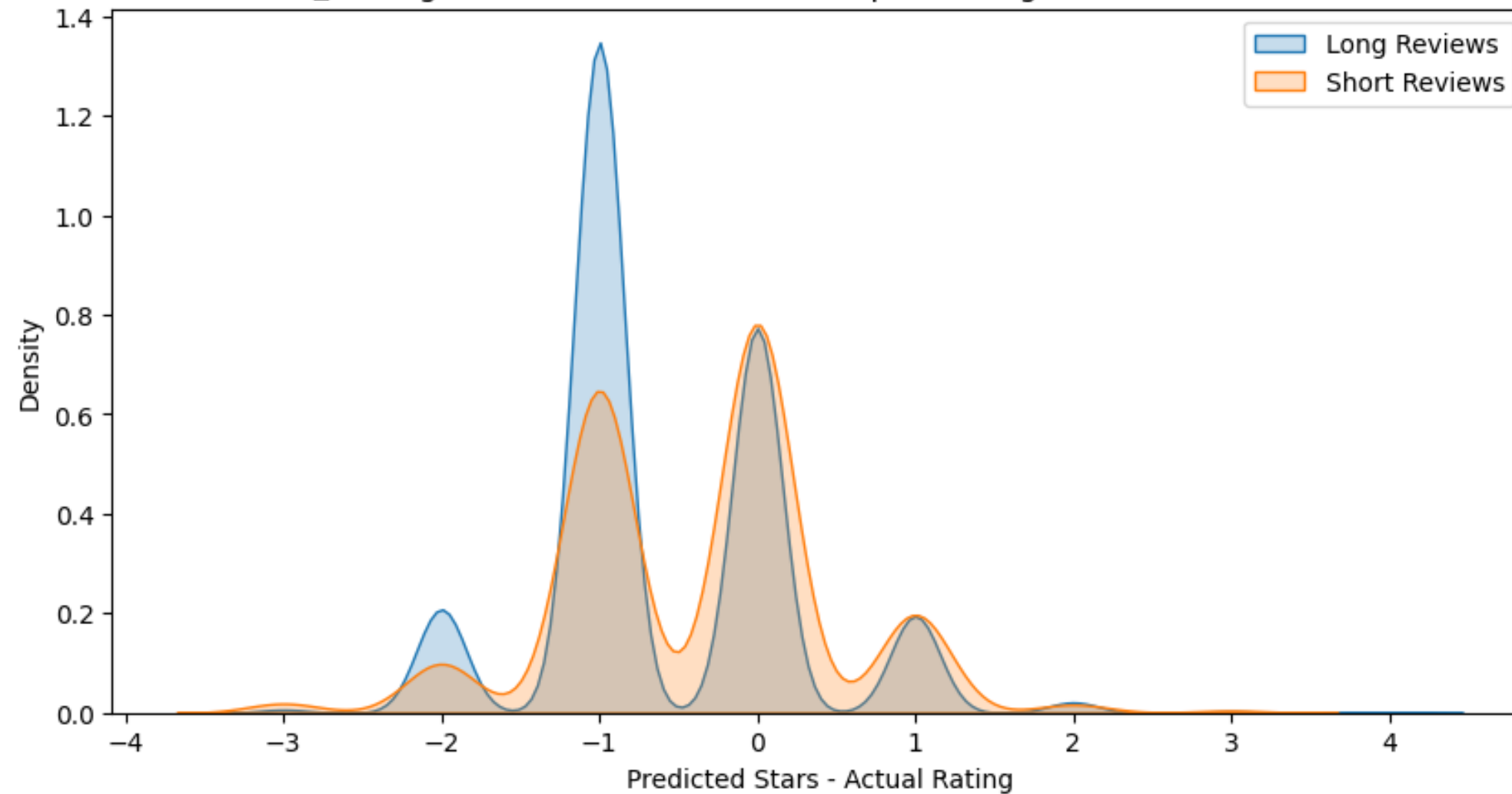
	Not Verified	Verified
topic		
0	19.110138	80.889862
1	58.190179	41.809821
2	11.490598	88.509402
3	3.653410	96.346590
4	21.337821	78.662179

Topic1:
Longest reviews, largest rating gap, lowest verified %
— investigate!

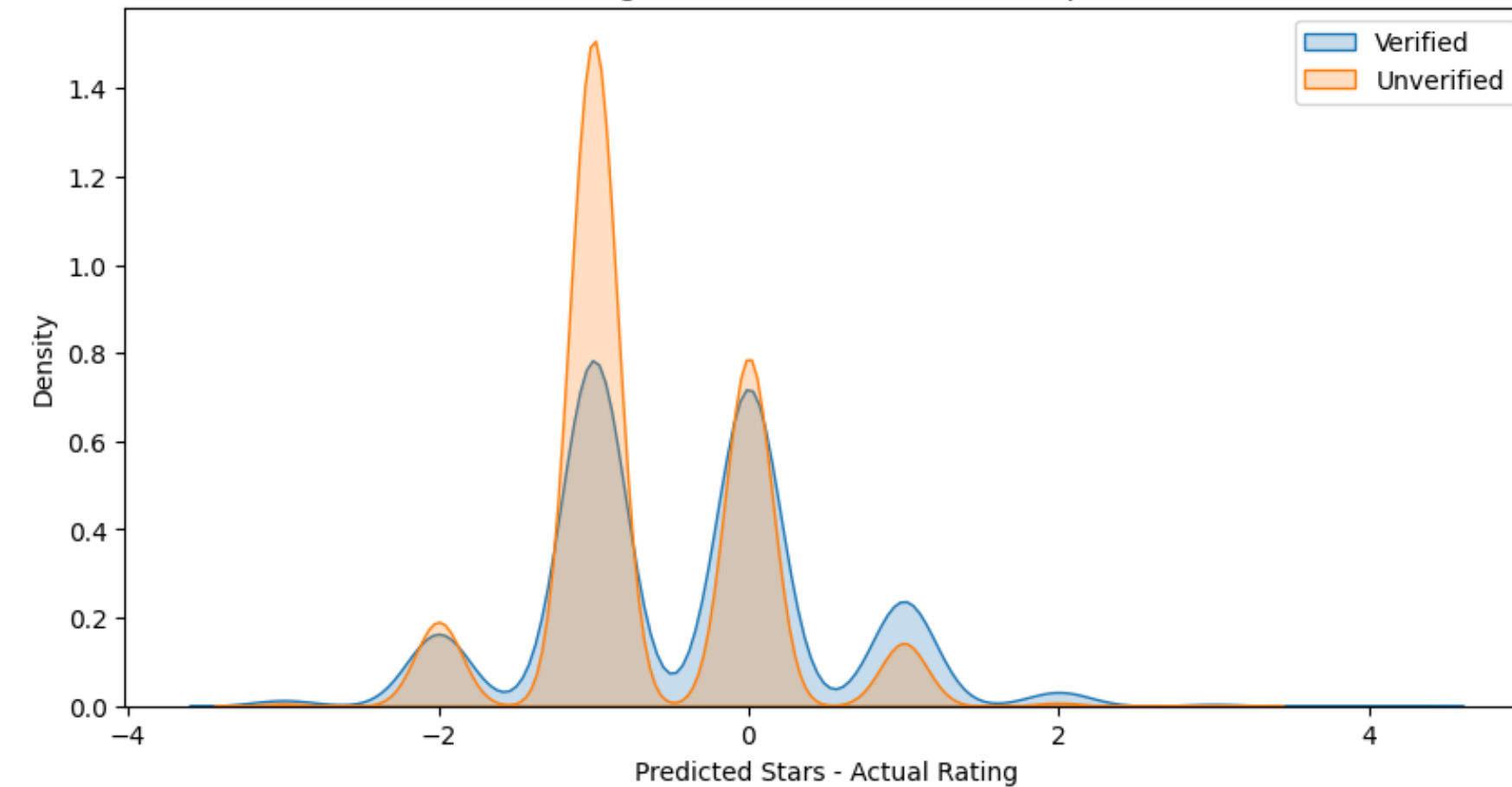


RQ2-1 Use TF-IDF vectorization and KMeans clustering ($k=5$)

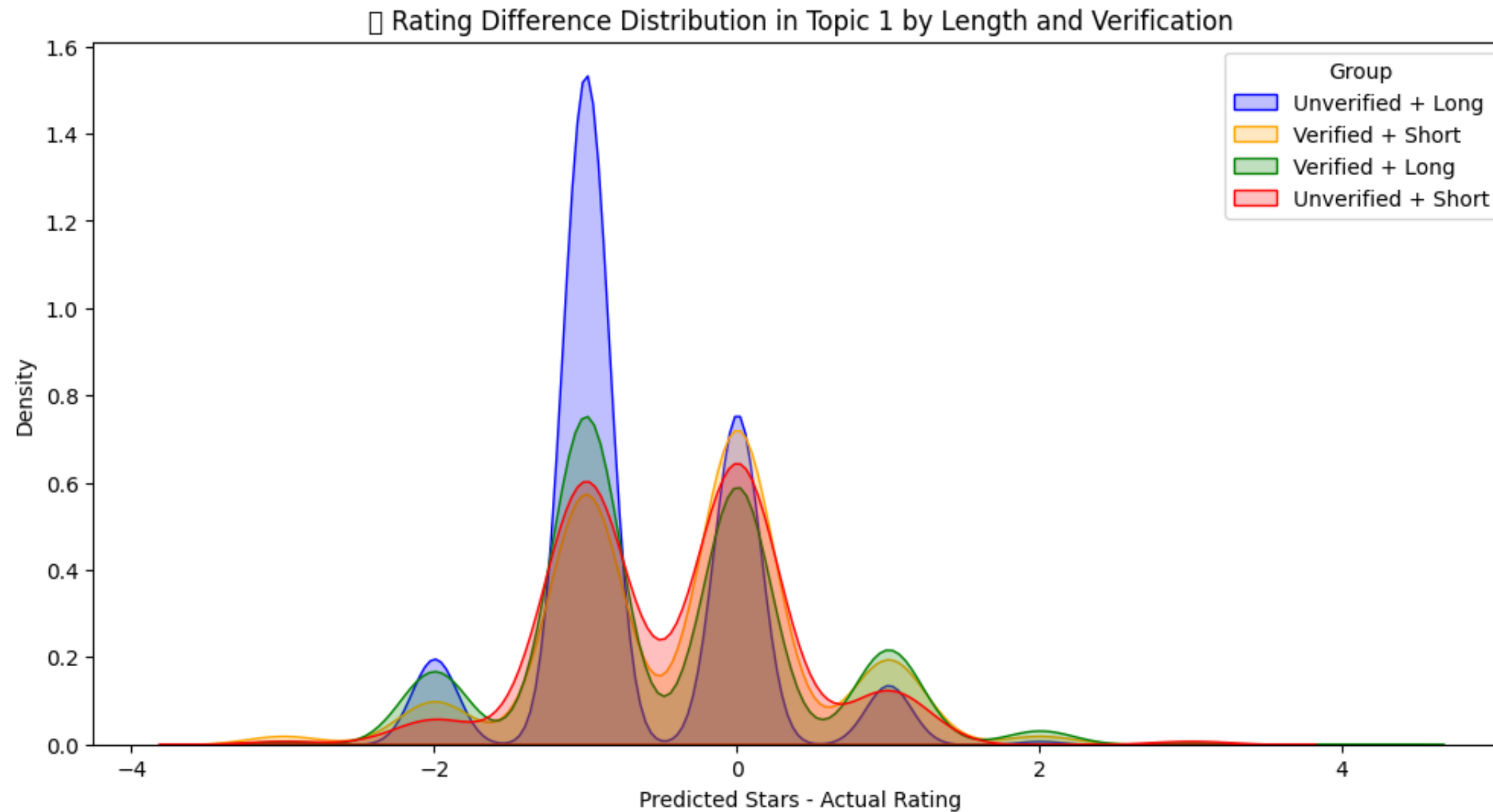
Rating Difference Distribution in Topic 1 (Long vs. Short Reviews)



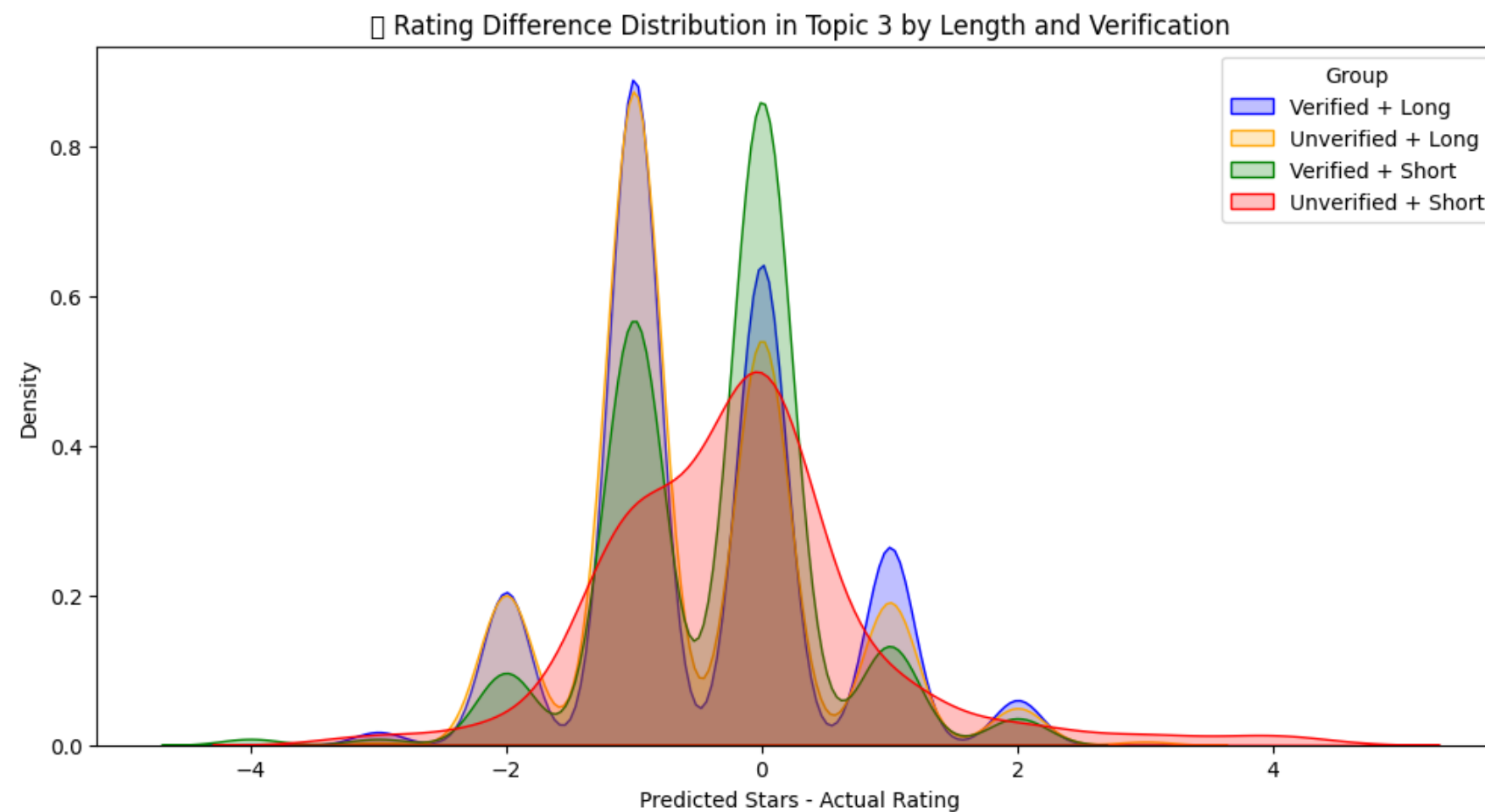
Rating Difference Distribution in Topic 1



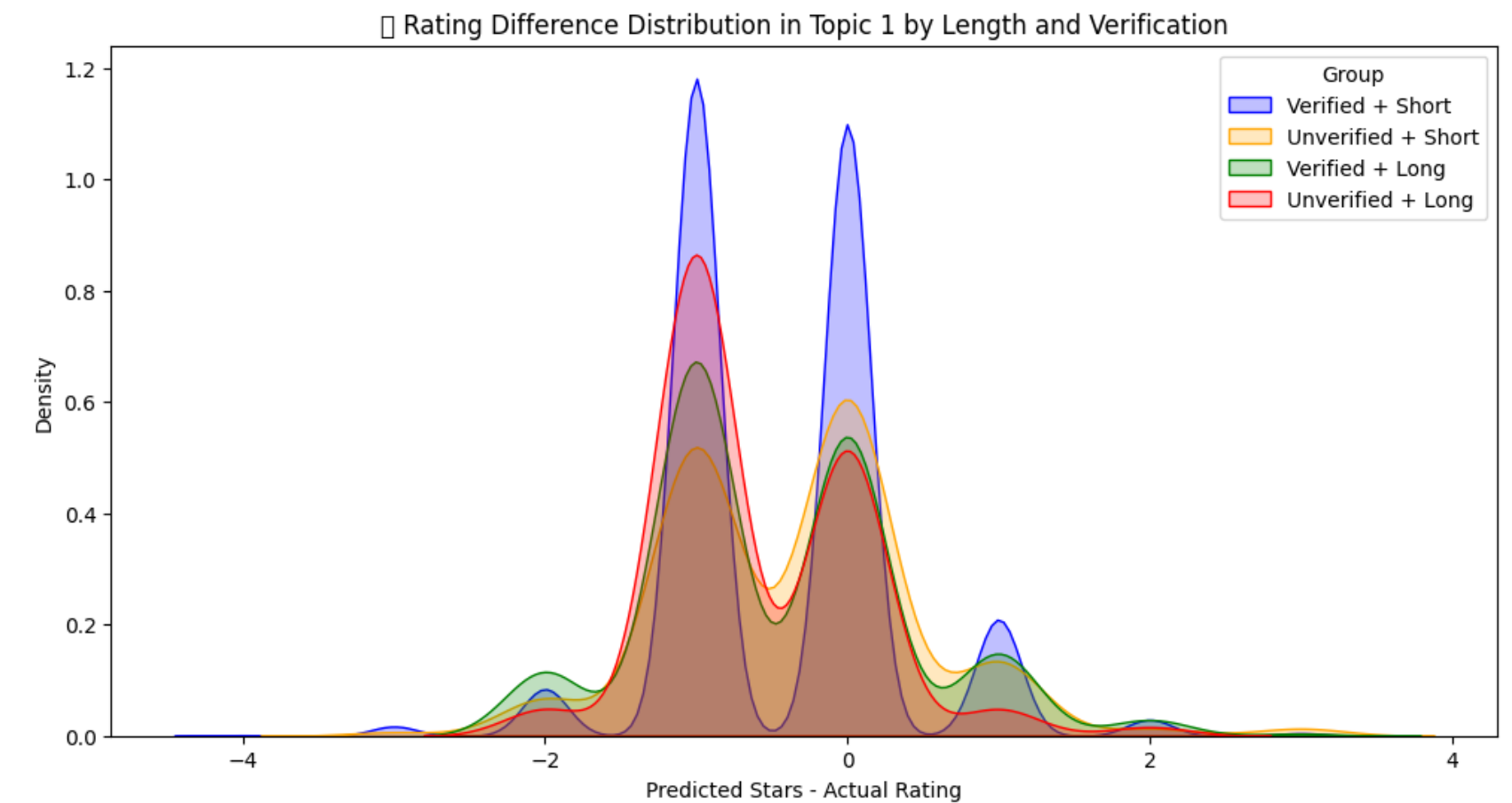
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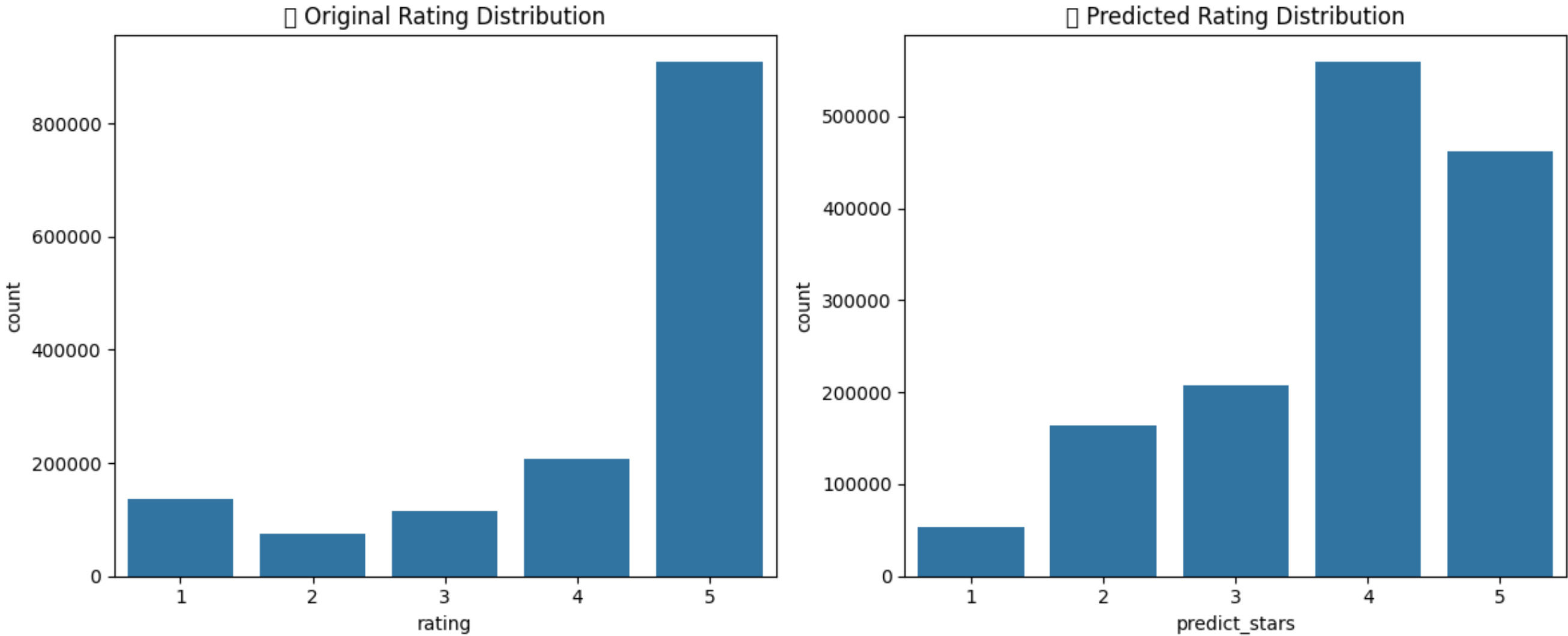
Electronics.csv



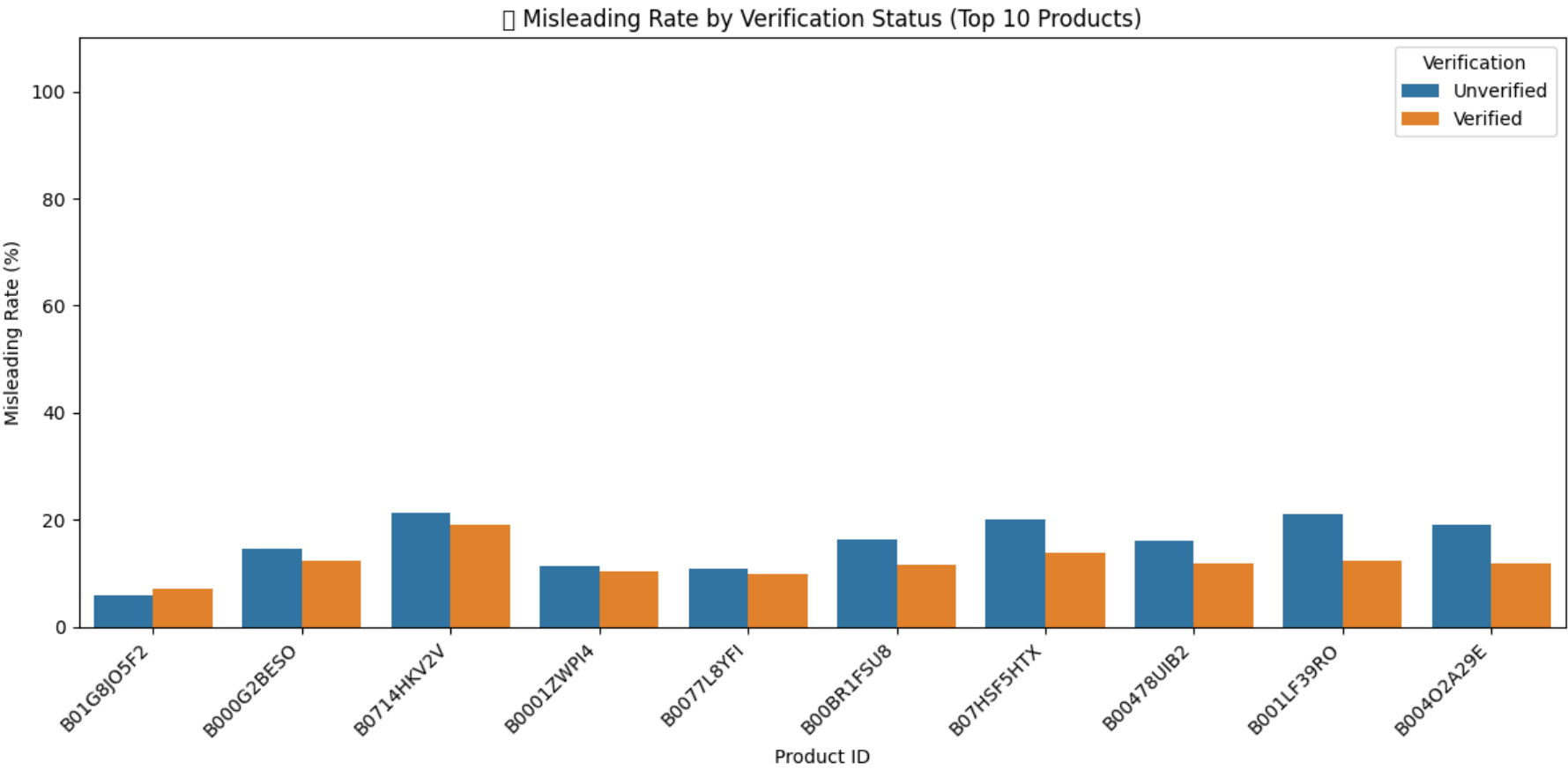
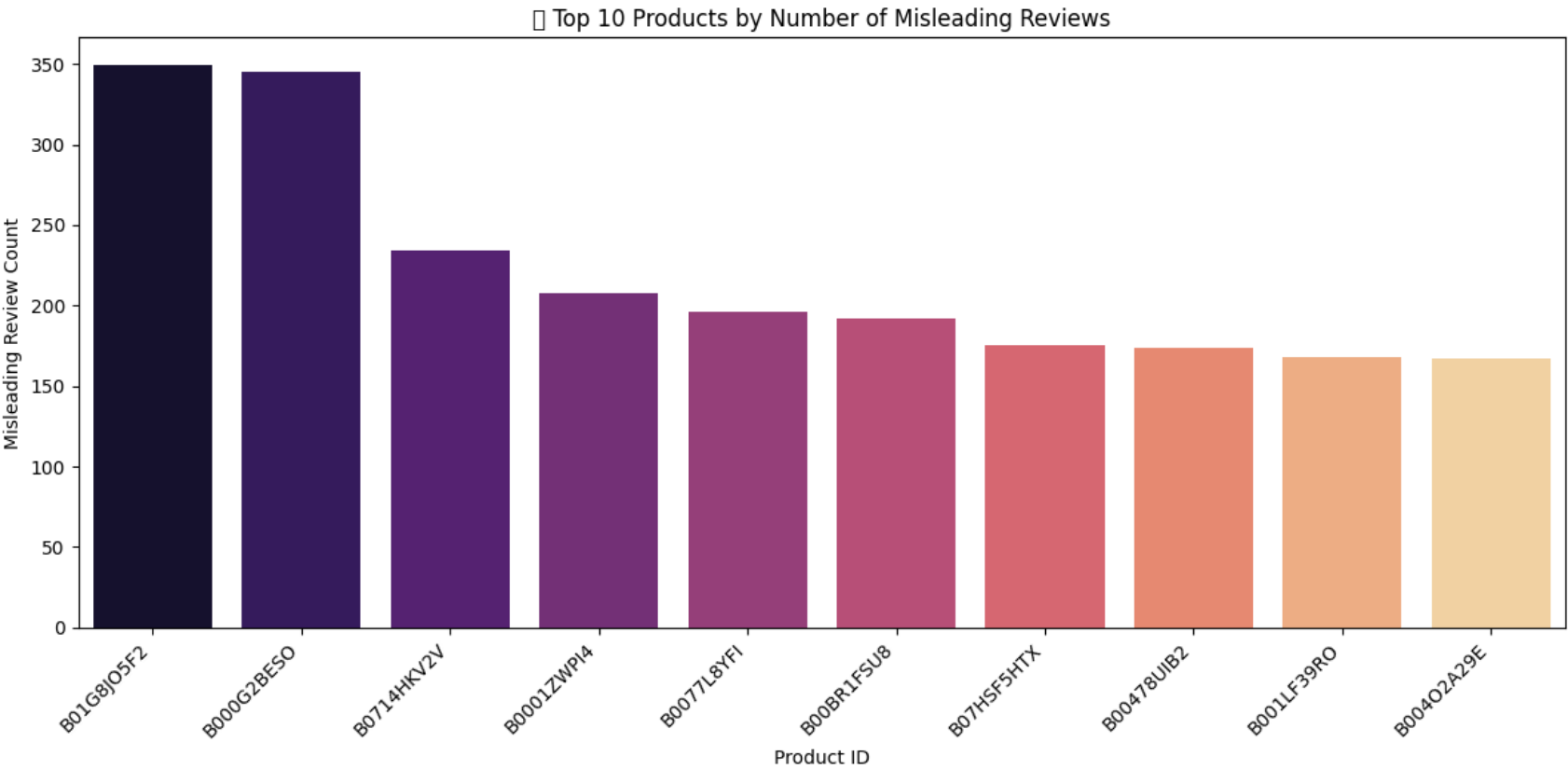
Health_and_Personal_Care.csv



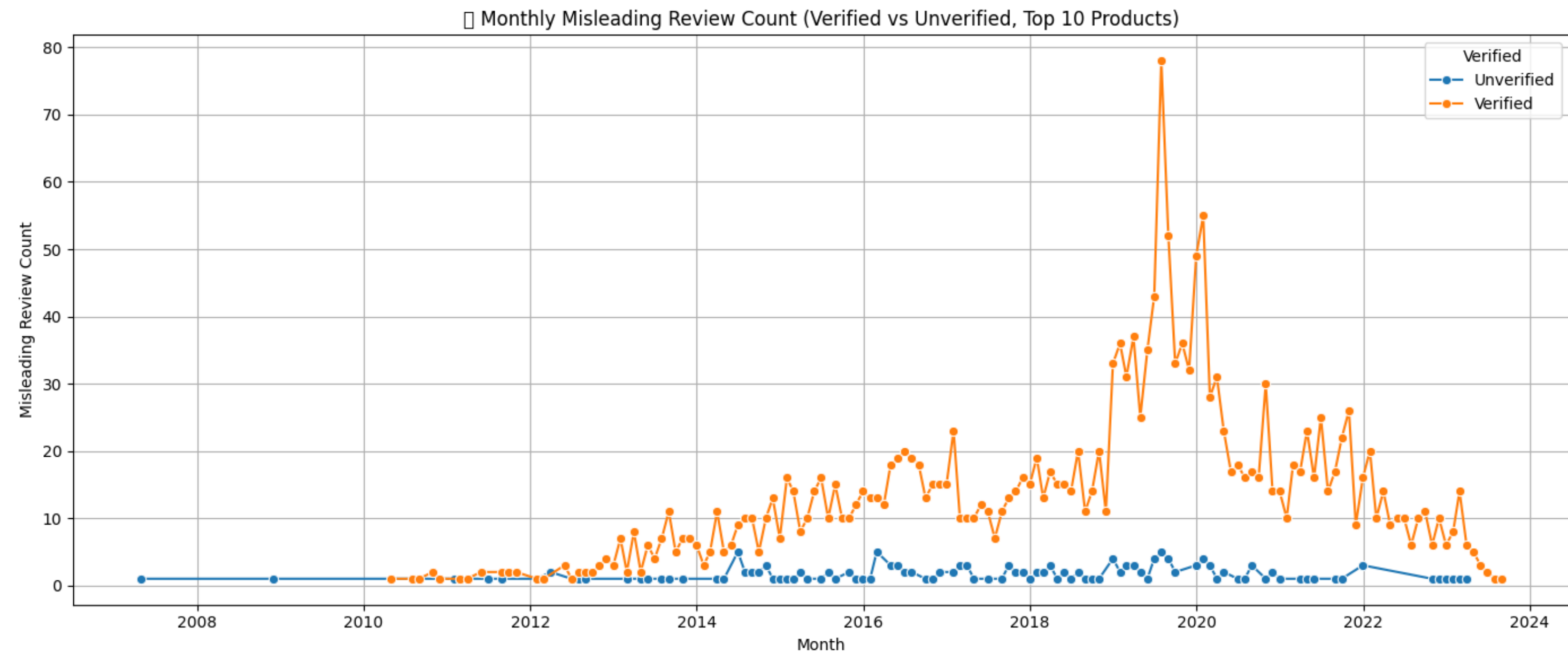
RQ2-2 Misleading instance on product and user level



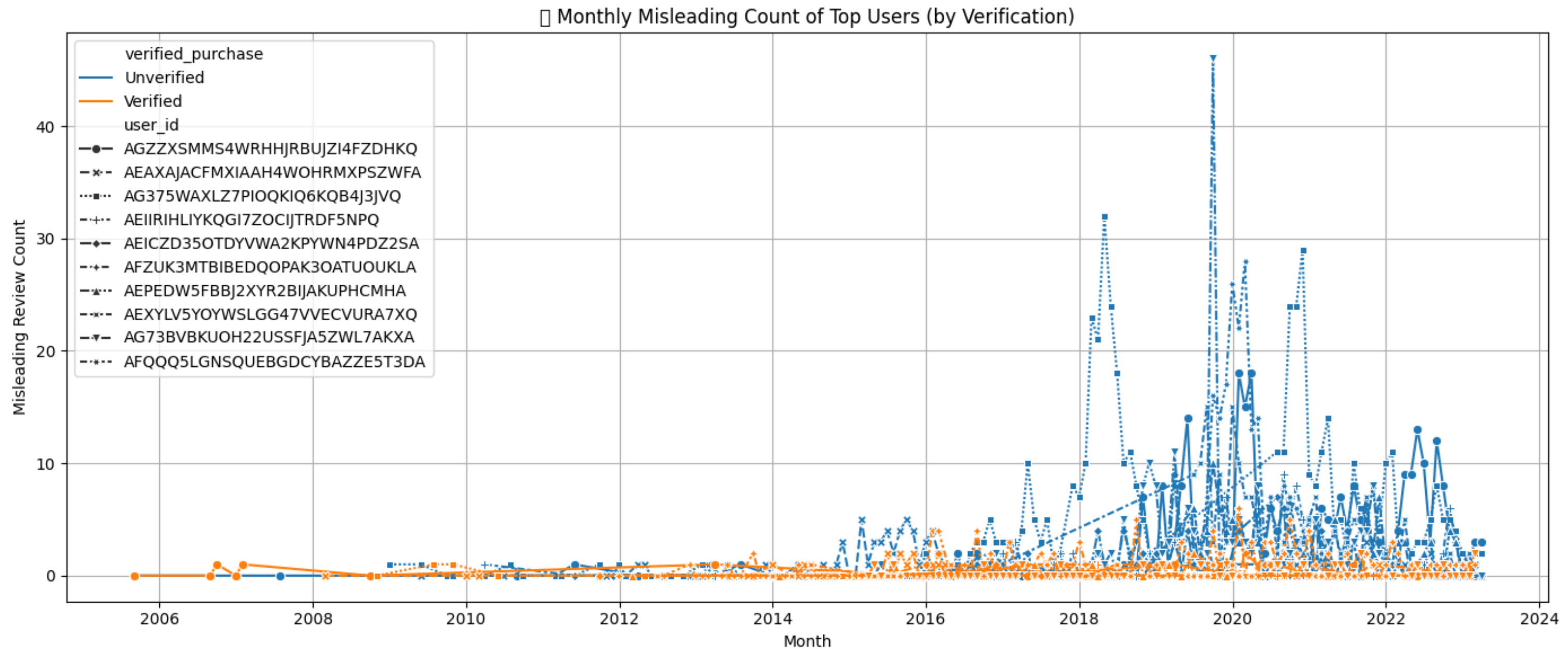
RQ2-2 Misleading instance on product and user level



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RQ2-2 Misleading instance on product and user level

- 1. Repeated Users Posting Misleading Reviews**
- 2. Long, Vague, or Contradictory Reviews**
- 3. Review Clusters in a Short Time Frame**
- 4. High Rate of Unverified Reviews**
- 5. Sentiment-Rating Mismatch**

Methodology of RQ3

Feature Extraction:

Sentiment: VADER scores for clean text.

Topics: LDA (5 topics) on vectorized text.

Formal: Rating, text length, image count, helpful votes, verified purchase.

Temporal: Time weight ($e^{(-\Delta t/365)}$).

Sentiment-Topic Analysis:

Step 1: BERT embeddings for negative reviews (10,000 samples), K-means clustering (k=5) for complaint themes.

Step 2: Reputation risk:

$$Risk = \left(\frac{TotalComments}{NegativeComments + 1} \right) \times |MeanSentiment|$$

Modeling:

Aggregated features by asin. Random Forest predicts sales proxy (log review count) and reputation risk, trained on 80:20 split.



Experiment results of RQ3

- Sales: $R^2 = 0.909$, RMSE = 0.176. Helpful votes (54.5%) and rating (20.7%) dominate (Table 1).
- Reputation: $R^2 = 0.772$, RMSE = 0.705. Sentiment (29.5%) and helpful votes (28.5%) lead.
- Complaint Themes BERT clusters reveal: poor quality
- (broken, cheap), sizing issues (small, fit), and general disappointment.
- Correlations Figure 1 shows low ratings and negative sentiment increase reputation risk.
- Figure 2 indicates high ratings and positive sentiment correlate with higher sales
- proxy (log review count).

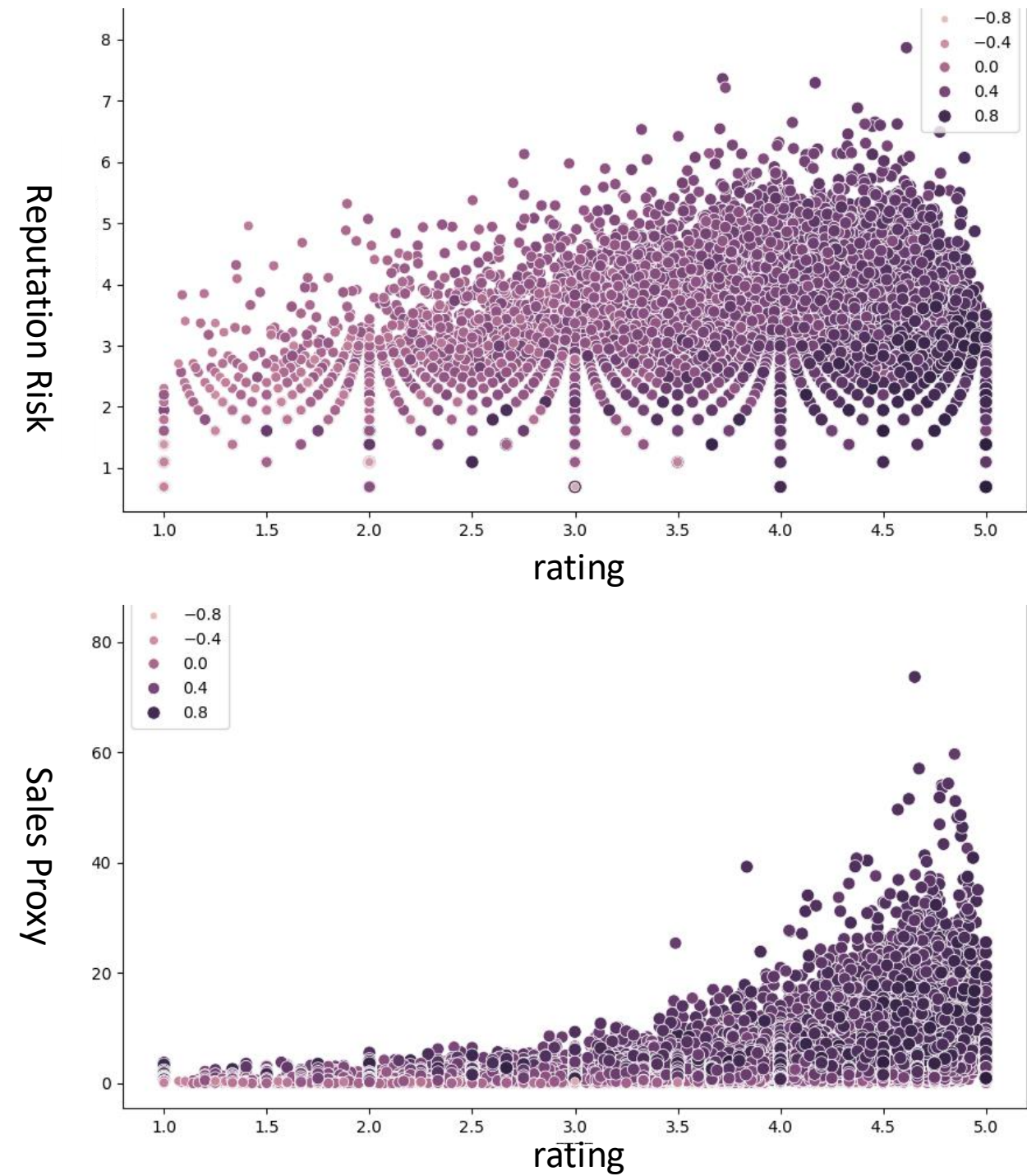


TABLE 1
Feature Importance

Feature	Sales	Reputation
helpful_vote	0.545	0.285
rating	0.207	0.192
sentiment	0.057	0.295

Conclusion

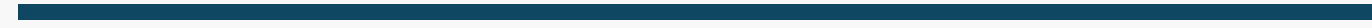
We studied sentiment–rating misalignment in Amazon product reviews using a fine-tuned BERT classifier.

Misalignment is common: many reviews show emotional tone that doesn't match the given star rating.

Key contributing factors include review length, punctuation use, and helpful votes. Temporal analysis showed that sentiment trends may shift over time, potentially affecting product reputation.

Text sentiment offers richer insight than ratings alone and can enhance review understanding.





Thank you

